QP CODE: 24900250

MAHATHMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS)

REGULAR EXAMINATION NOVEMBER 2024

First Semester

Discipline Specific Core Course: MG1DSCSAC100

INTRODUCTION TO STRENGTH & CONDITIONING

(2024 ADMISSION ONWARDS)

SCHEME OF EVALUATION

Duration: 1.5 Hours **Maximum Marks: 50**

Part A (Multiple choice questions)

Answer all questions.

Each question carries 1 mark.

1. Tudor Bompa
2. Functional capacity
3. Overload
4. Incorporate recovery phases
5. Overuse of certain muscle groups
6. Releasing muscle tightness and improving tissue quality
7. Reducing muscle soreness, inflammation, and swelling
8. It prevents overuse of the injured area while maintaining overall fitness
9. Around the organs
10. Plyometric

Part B (Short answer questions)

Answer 10 questions.

Each question carries 2 marks.

11. How has technology impacted strength and conditioning in recent years?
    In recent years, technology has greatly impacted strength and conditioning by enabling precise tracking of performance metrics, such as force, speed, and recovery rates. Wearable devices, apps, and data analytics allow coaches to tailor training, monitor progress in real time, and make evidence-based adjustments to optimize athlete performance and minimize injury risk.
12. What is the role of a strength and conditioning coach?
    The role of a strength and conditioning coach is to design and implement training programs that enhance athletes' physical performance, focusing on strength, power, speed, agility, and endurance. They also work to reduce injury risk, promote proper recovery, and support athletes' overall fitness to meet the demands of their specific sports.
13. Why is strength and conditioning considered essential for athletes?
    Strength and conditioning is essential for athletes because it improves physical performance by developing strength, power, speed, and endurance, while also reducing the risk of injuries. Tailored training programs help athletes enhance their sport-specific abilities, recover faster, and perform at their highest potential during competition.

14. What is muscular imbalance, and how can it contribute to injury?
    Muscular imbalance occurs when one muscle or muscle group is significantly stronger or tighter than its opposing muscle group. This imbalance can lead to improper movement patterns, placing excessive stress on joints and other tissues. Over time, it increases the risk of injury, as the body compensates for the weakness or tightness, potentially causing strains, sprains, or joint instability.
15. Explain the role of mobility exercises in a rehabilitation program for athletes.
    Mobility exercises in rehabilitation help restore range of motion, improve joint function, and rebuild strength and control in injured areas. They gradually prepare athletes to return to full activity by enhancing flexibility, reducing stiffness, and preventing compensatory movement patterns that could lead to re-injury.
16. Why is sports-specific rehabilitation important for an athlete's return to play?
    Sports-specific rehabilitation is important for an athlete's return to play because it tailors recovery exercises to the movements and demands of their sport. This ensures the athlete regains the strength, coordination, and skills required for optimal performance while minimizing the risk of re-injury by addressing sport-specific motions and stresses.
17. What specific cardiovascular benefits arise from strength training?
    Strength training provides several cardiovascular benefits, including improved heart efficiency, increased circulation, and better blood flow. It helps lower resting heart rate and blood pressure over time. Additionally, it enhances vascular health by promoting the development of capillaries, improving oxygen delivery to muscles, and supporting overall cardiovascular endurance when combined with aerobic activity. Strength training also contributes to weight management, which reduces the risk of heart disease.
18. How can nutrition influence body composition?
    Nutrition significantly influences body composition by affecting fat loss, muscle gain, and overall energy balance. Adequate protein intake supports muscle growth and repair, while controlling caloric intake helps reduce body fat. Balanced macronutrients (carbohydrates, proteins, and fats) and proper nutrient timing can optimize muscle preservation and fat metabolism, helping athletes achieve a leaner, more muscular body composition.
19. What are the stages of Long-Term Athlete Development (LTAD)?
    - Active Start
    - FUNdamentals
    - Learn to Train
    - Train to Train
    - Train to Compete
    - Train to Win
    - Active for Life
20. What factors can influence the effectiveness of skill integration practices in an athlete's training program?
    Individual Skill Level: Athletes with varying levels of technical proficiency may require different approaches to integrate skills effectively.
    Sport-Specific Demands: The nature of the sport, including its intensity, pace, and complexity, will dictate the types of skills integrated during training.
    Training Environment: The availability of resources, such as equipment or space, and the quality of coaching support, directly impact how effectively skills can be integrated into training sessions.
    Recovery and Fatigue: Overtraining or inadequate recovery can hinder skill execution and the ability to apply integrated skills under pressure.
21. Compare and contrast the adaptations seen in a sprinter versus a marathon runner based on the SAID principle.

Based on the SAID principle, a sprinter develops fast-twitch muscle fibers, improving explosive power and speed through high-intensity, short-duration training. In contrast, a marathon runner adapts by enhancing slow-twitch muscle fibers, increasing endurance and aerobic capacity through long-duration, low-intensity training. While both adapt to their specific demands, the sprinter prioritizes power and speed, and the marathon runner focuses on stamina and energy efficiency.

22. What role does a coach plays in Performance Monitoring?

Collecting and Analysing Data: Coaches track and interpret performance metrics, such as strength, speed, and endurance, to assess athlete progress and adjust training programs accordingly.

Providing Feedback and Adjustments: Based on the data, the coach offers constructive feedback to the athlete, making necessary adjustments to the training regimen to optimize performance and address any identified weaknesses or risks.

Part C (Essay type questions)

Answer 4 questions.

Each question carries 5 marks.

23. Discuss the need for and importance of strength and conditioning in modern sports.

- **Improves Athletic Performance**: Enhances physical attributes like strength, power, speed, and agility, which are crucial for success in competitive sports.
- **Injury Prevention**: Strengthening muscles, ligaments, and tendons reduces the risk of injuries and ensures better joint stability.
- **Sport-Specific Training**: Tailored conditioning programs address the unique demands of each sport, improving skills and overall efficiency.
- **Enhances Recovery**: Proper conditioning improves recovery times by promoting circulation and muscle repair, allowing athletes to train harder and more frequently.
- **Mental Toughness and Confidence**: A well-structured program builds resilience and boosts confidence, preparing athletes for high-pressure situations in competition.

24. Analyze the impact of age and gender on flexibility and mobility.

Impact of Age

- **Decrease in Flexibility**: Flexibility declines with age due to reduced elasticity of muscles, tendons, and ligaments.
- **Joint Stiffness**: Aging leads to loss of cartilage and synovial fluid, affecting mobility.

Impact of Gender

- **Women Tend to Be More Flexible**: Due to hormonal factors (e.g., estrogen) and differences in joint structure, women generally exhibit greater flexibility than men.
- **Gender-Specific Activities**: Females often engage in flexibility-focused activities like dance and yoga, further enhancing mobility.

Combined Effects: Older women may retain more flexibility compared to men of the same age, but aging impacts both genders similarly in terms of overall mobility loss.

25. How does technology (e.g., wearable devices, rehabilitation software) impact the rehabilitation process in sports?

- **Accurate Assessment and Monitoring**: Wearable devices track metrics like range of motion, muscle activity, and progress in real time, enabling precise evaluation.
- **Personalized Rehabilitation Plans**: Rehabilitation software uses data to create customized recovery programs tailored to individual needs and injury severity.
- **Enhanced Feedback Mechanisms**: Real-time feedback from devices helps athletes correct movements, ensuring proper technique during recovery exercises.
- **Motivation and Engagement**: Gamification elements in rehabilitation apps keep athletes motivated and consistent in following recovery routines.
- **Remote Rehabilitation**: Tele-rehabilitation technologies enable virtual sessions, making therapy accessible and convenient for injured athletes.

26. What are the key components of a healthy lifestyle and how do they contribute to overall well-being?

- **Balanced Diet**: Provides essential nutrients for energy, growth, and repair, promoting physical health and preventing diseases.
- **Regular Physical Activity**: Enhances cardiovascular health, strengthens muscles, and boosts mental well-being by releasing endorphins.
- **Adequate Sleep**: Supports recovery, improves cognitive function, and regulates hormones critical for health.
- **Stress Management**: Reduces mental strain, enhances emotional stability, and prevents chronic conditions like hypertension.
- **Avoidance of Harmful Habits**: Avoiding smoking, excessive alcohol, and drug use safeguards against various lifestyle diseases and improves quality of life.

27. What is the Long-Term Athlete Development (LTAD) model, and what is it's significance?

- **Definition of LTAD**: The LTAD model is a framework that outlines the optimal development of athletes across different age groups and stages, focusing on physical, psychological, and technical growth.
- **Stages of LTAD**: Active Start, FUNdamentals, Train to Train, Train to Compete, and Train to Win.
- **Importance of LTAD**
    - **Skill Development**: Ensures proper skill acquisition at the right stages of growth.
    - **Injury Prevention**: Promotes age-appropriate training to avoid overtraining and injuries.
    - **Lifetime Participation**: Encourages lifelong involvement in physical activity, even after competitive careers.
    - **Athlete Optimization**: Helps athletes reach their full potential by addressing physical, mental, and tactical aspects systematically.
- **Impact on Sports**: Provides a structured pathway for talent identification, nurturing, and achieving peak performance.

28. What is the primary role of a strength and conditioning coach in an athlete's training regimen?

- **Developing Customized Training Program:** Designs sport-specific strength, power, and endurance programs tailored to the athlete's needs and goals.
- **Enhancing Athletic Performance**: Focuses on improving physical attributes like speed, agility, flexibility, and explosive power to maximize performance.
- **Injury Prevention**: Implements exercises to strengthen muscles, joints, and connective tissues, reducing injury risks.
- **Monitoring Progress**: Tracks performance metrics and adjusts training plans to ensure continuous improvement and prevent overtraining.
- **Building Mental and Physical Resilience**: Helps athletes develop discipline, confidence, and the ability to perform under pressure through structured conditioning.